FactorJEPA: World Models That Predict Crowded, Chaotic City Streets

Anonymous submission

Abstract

World models capture the structure and dynamics of the physical world; Joint Embedding Predictive Architectures (JEPA) are a compelling direction. We study a largely unexplored regime: crowded, chaotic Global South city streets (**DENSEWORLD**), where soft boundaries, agent heterogeneity, occlusion, and social negotiation replace lane-structured assumptions. We release DENSEWORLD 1.0: 115,687 videos (276 hours) across 22 Indian cities, 86,831 with automatically derived *layout*, *agent*, and *interaction* targets. We introduce ***FactorJEPA***, which makes world structure a first-class predictive primitive via *predictor surgery*: a three-stage curriculum over separated layout, agent, and interaction views. Against the frozen V-JEPA 2.1 backbone (1B) it cuts future-frame L1 by 10.6%, causal future-block L1 by 8.9%, and mask-ratio slope by 27.4%, and raises motion-cosine separation to 15.3× the frozen value, each gain exceeding 70× the paired 95% CI. Against all six fine-tuning baselines (full FT, LP-FT, LoRA, DoRA, Auto-RGN, continual-SSL) it separates on both prediction diagnostics at 2B and 1B (24 of 24 paired wins), beating the strongest, LoRA, by +2.8%, +1.6%, +25.2%, and +17.8%. Rankings replicate across scale ($\rho = 0.895$ to 0.978), making the 1B model a half-cost proxy for 2B. We release the pipeline, checkpoints, and 15-diagnostic harness.

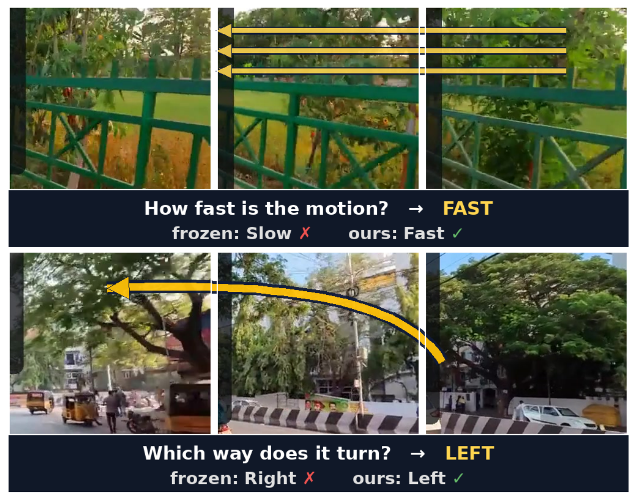

Figure 1: **Same question, two encoders.** Each row is a three-frame filmstrip of one held-out clip posed as a multiple-choice question, answered from an identical probe head over the frozen V-JEPA 2.1 encoder vs. ours (factor-view predictor surgery); only the backbone differs. *Top:* motion speed (ours 69.8% vs frozen 60.9%). *Bottom:* turn direction.

DENSEWORLD: A Benchmark for Populous, Crowded, and Chaotic Global South

We use DENSEWORLD to denote a critical but underrepresented world-modeling regime: populous, crowded, and chaotic Global South urban scenes. It is defined not only by geography, but by measurable properties: high agent count density, high agent occupancy, persistent occlusion, agent heterogeneity, and interaction pressure. This regime matters because many emerging urban environments combine high population density, mixed mobility systems, diverse infrastructure patterns, and rapid spatial change, making them scientifically important yet underrepresented in world-model benchmarks (Mahendra et al. 2021; Ellis and Roberts 2016). Thus, world models must handle uncertainty, density, and relational complexity that is less prominent in lower-density, lane-structured benchmarks. See Figure 2 for representative DENSEWORLD scenes.

We characterize DENSEWORLD through four coupled properties. First, it exhibits soft spatial boundaries: road edges, drivable space, and pedestrian regions are often negotiated rather than crisply marked (Varma et al. 2019; Dokania et al. 2023). Second, it contains extreme agent heterogeneity, with pedestrians, cars, two-wheelers, carts, animals, delivery vehicles, and rickshaws coexist locally (Khan and Maini 1999; Asaithambi et al. 2012). Third, it has persistent occlusion from density, clutter, and near-field encounters, producing severe partial observability (Varma et al. 2019; Dokania et al. 2023); prediction must rely on *memory*, *object permanence*, and *relational inference*, not only visible appearance. Fourth, motion is governed less by rigid rule compliance than by rapid social negotiation under mixed traffic and weak lane discipline (Papathanasopoulou and Antoniou 2018; Asaithambi et al. 2012).

Together, these properties make DENSEWORLD a natural stress test for world models: success requires representations that preserve layout, agent state, and interaction dynamics under uncertainty, rather than collapsing into one entangled predictive latent. As detailed below, we measure this

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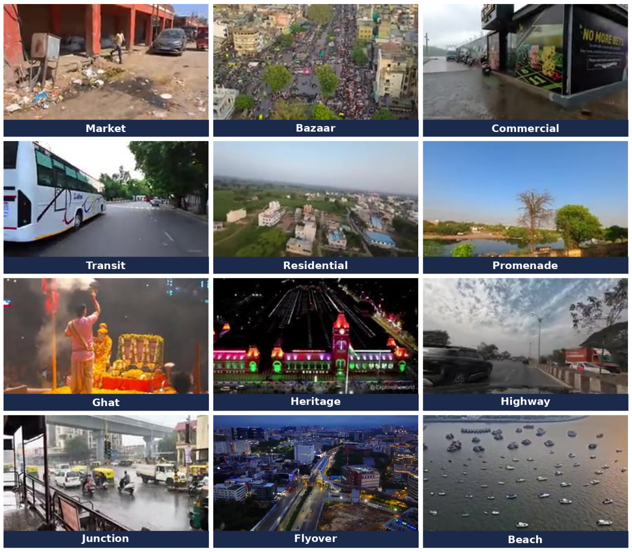

Figure 2: **DENSEWORLD 1.0 scene-type coverage.** Twelve representative scene types (market, bazaar, commercial, transit, residential, promenade, ghat, heritage, highway, junction, flyover, beach) across 22 Indian cities.

regime through agent count density, agent occupancy, occlusion pressure, interaction pressure, and agent heterogeneity. This motivates a predictor that separates layout, entities, and interactions.

DENSEWORLD 1.0: the Benchmark

We collected 714 long-form YouTube videos at 480p, totalling 276 hours of urban footage across 22 Tier-1 and Tier-2 cities in India. The data spans drive-through, walk-through, and aerial viewpoints, and long-form recordings are segmented using scene-detection and shot-boundary methods into 115,687 short videos of 4 to 10 seconds, of which 86,831 carry automatically derived layout, agent, and interaction targets. DENSEWORLD 1.0 covers commercial, residential, transit, heritage, coastal, and high-density street settings, including markets, commercial streets, transit corridors, junctions, flyovers, bazaars, ghats, beaches, and skylines. It further spans variations in the time of day, weather, crowd density, traffic mix, pedestrian–vehicle separation, road geometry, surface quality, encroachment, and lighting. Beyond standard urban actors, the dataset includes locally frequent agents and objects such as auto-rickshaws, cycle rickshaws, street vendors, and animals. Before release, all footage is processed through privacy-preserving filters, including face and license-plate blurring and removal of sensitive segments.

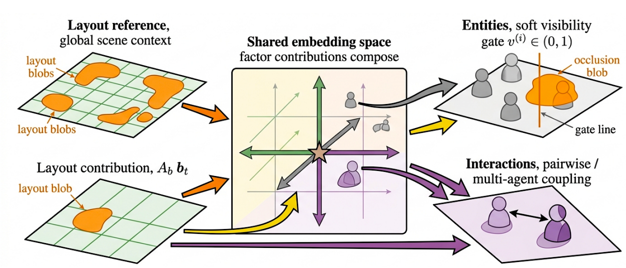

Figure 3: ***Factorized predictive channels.*** Context is split into *layout*, visibility-gated *agent*, and sparse *interaction* coordinates, then recomposed in the future JEPA embedding, limiting cross-factor leakage.

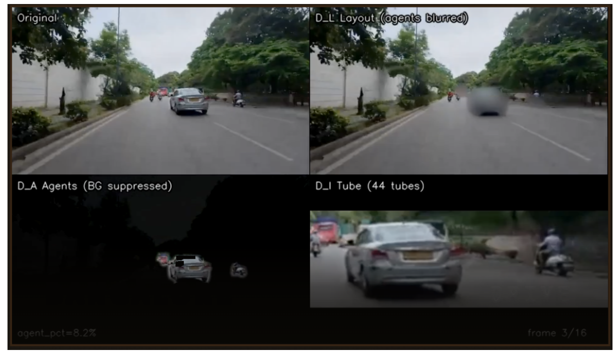

Figure 4: ***DINO-based factor-target construction.*** Per clip, the pipeline derives *layout* D_L, *agent* D_A, and *interaction* D_I supervision by masking foreground/background and linking agent tubes.

A defining feature of DENSEWORLD is heterogeneous mixed traffic. In many developing-country cities, multiple vehicle and actor types share the same right of way under weak lane discipline, with strong lateral and longitudinal coupling (Khan and Maini 1999; Papathanasopoulou and Antoniou 2018; Asaithambi et al. 2012). This is not a mild variant of standard traffic, but a qualitatively different operating regime. Prior work highlights the coexistence of cars, buses, trucks, auto-rickshaws, scooters, motorcycles, bicycles, carts, and other slow- and fast-moving agents within the same corridor (Khan and Maini 1999; Asaithambi et al. 2012). Here, scene structure does not decompose into *static lanes*, *isolated actors*, and *simple pairwise relations*; spatial support is fluid, visibility is unstable, and behavior is shaped by dense local negotiation.

How Dense is DENSEWORLD? Comparison with Standard Driving Benchmarks

To make this regime measurable, we compare DENSEWORLD against BDD100K and nuScenes, two widely used benchmarks for autonomous-driving perception and scene understanding (Yu et al. 2020; Caesar et al. 2020). The goal is not to claim that DENSEWORLD scenes merely look different, but to test whether they occupy a quantitatively distinct operating regime for world-model evaluation.

We measure this regime along five axes: agent count density, agent occupancy, occlusion pressure, interaction pressure, and agent heterogeneity. These metrics capture, respectively, the number of dynamic agents, the fraction of image area they occupy, the frequency of partial or heavy occlusion, the density of spatially proximate agent pairs, and the diversity of co-occurring actor categories. Together, they quantify the multi-agent load, visual congestion, visibility degradation, local interaction structure, and traffic diversity of each scene.

For an auditable comparison, all statistics are computed after mapping labels to a shared dynamic-agent taxonomy and matching comparable scene categories across datasets. Figure 5 shows that DENSEWORLD has consistently higher density and occupancy across matched categories, with the largest gaps in market, commercial, and flyover/underpass scenes. These results support the central premise of the benchmark: DENSEWORLD is not merely a geographic extension of existing driving data; it defines a higher-density, higher-occlusion, and higher-interaction regime for evaluating predictive world models. Full automatic-annotation details, including thresholds, annotation taxonomy, export format, and loss mapping, are provided in Appendix C.

Can Fine-Tuning Close the DENSEWORLD Gap?

Before introducing FactorJEPA, we ask whether conventional adaptation can recover the predictive structure required by DENSEWORLD without reorganizing the JEPA predictor. We evaluate three complementary strategies: LoRA (Hu et al. 2022), DoRA (Liu et al. 2024b), and Auto-RGN (Lee et al. 2023). All use the same raw clips, optimization steps, optimizer, masking policy, and evaluation protocol; only the parameter-update mechanism changes.

Let x denote a video clip; M_c and M_t the context and target masks; f_θ the online encoder; $\bar{f}_{\bar{\theta}}$ the momentum target encoder; and g_ϕ the predictor. All baselines retain the executed V-JEPA objective (Bardes et al. 2024) $\mathcal{L}_{\text{JEPA}} = \mathbb{E}_x \big\| g_\phi(f_\theta(M_c \odot x), M_t) - \text{sg}\ \bar{f}_{\bar{\theta}}(M_t \odot x) \big\|_2^2$, where sg is stop-gradient. This data- and step-matched protocol isolates adaptation capacity without introducing a different prediction target or training signal.

Formal update rules for LoRA, DoRA, and Auto-RGN, their protocol-matched configuration, and per-baseline ranks, block budgets, trainable-parameter counts, and measured GPU-hours are given in Appendix C.

FactorJEPA: Predictor Surgery over Factorized Views

Motivation. Let x be a video clip, M_c and M_t its context and target masks, f_θ the online encoder, and $\bar{f}_{\bar{\theta}}$ the momentum target encoder. A conventional JEPA predictor g_ϕ learns

$$g_\phi : (f_\theta(M_c \odot x), M_t) \longmapsto \widehat{Y}_{t+\Delta}, \qquad Y^\star_{t+\Delta} = \text{sg}\big[\Pi_{M_t} \bar{f}_{\bar{\theta}}(x)\big] \,,$$

by matching $\widehat{Y}_{t+\Delta}$ to $Y^\star_{t+\Delta}$. The objective specifies *what* to predict, but every clip presents layout, agents and interactions simultaneously, so nothing constrains *which* of them the predictor actually uses.

This ambiguity is consequential in DENSEWORLD, where layout, agent density, visibility and interaction pressure are strongly correlated. A high-capacity predictor can exploit *shortcut mixtures* (crowd texture as a proxy for interaction, visible appearance for object persistence, or road geometry for motion) without recovering the structure governing scene evolution.

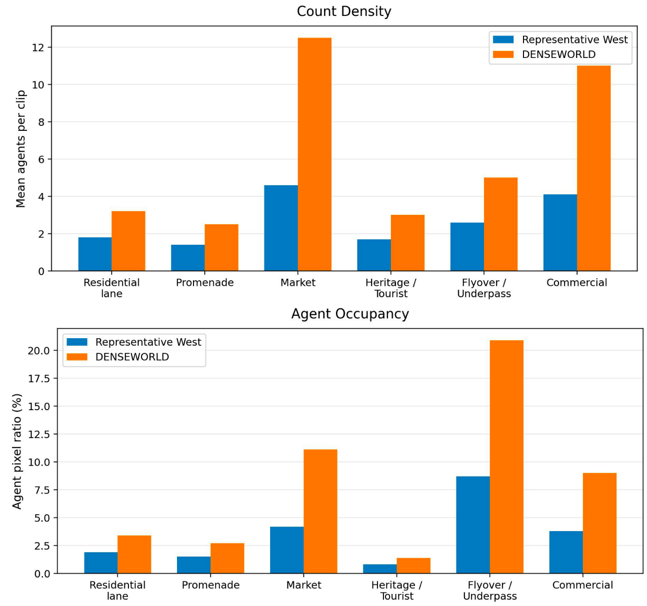

Figure 5: **DENSEWORLD exceeds standard driving benchmarks in multi-agent density and occupancy**, with the largest gaps in market, commercial, and flyover scenes.

FactorJEPA addresses this on the *input* side rather than by redesigning the predictor. We construct three complementary views of every clip, in which layout, agents and interactions are isolated by construction, and adapt a pretrained V-JEPA predictor over them as a staged curriculum. The architecture is unchanged; what changes is the sequence of distributions the predictor is asked to model, and how much of the encoder is free to move at each stage.

Automatic Factor-View Construction

For each privacy-filtered clip x_n we run an open-vocabulary detector (Grounding DINO, Swin-B) (Liu et al. 2024a) to localise dynamic agents, and propagate those detections through the clip with a promptable video segmenter (SAM 3) (Carion et al. 2025). Detections are re-seeded at four anchor frames and propagated within each anchor's segment, which keeps identities stable under the heavy occlusion typical of the regime. This yields, per clip, a set of agent masks and temporally associated tracklets (a mask and box sequence per agent over its lifespan; formal notation in Appendix C).

A deterministic map converts these into three *views* of the same clip, each a video the predictor can be trained on directly:

- D_L (layout): agent regions are suppressed, leaving slowly varying spatial support: road geometry, buildings, drivable area.
- D_A (agents): background context is suppressed, leaving the dynamic actors and their motion.
- D_I (interactions): spatio-temporal tubes around co-occurring, spatially proximate tracklet pairs, isolating local multi-agent coupling.

Views are derived only from the training split; no detector or segmenter output is used as an evaluation label. Checkpoints, prompt vocabulary, association rules, thresholds and per-view coverage are reported in Appendix C.

Three-Stage Curriculum

Training proceeds over the three views in order of increasing relational complexity, $D_L \rightarrow D_A \rightarrow D_I$, with a mixture weight over views at each stage. The ordering is deliberate: layout is the most stationary signal and anchors the encoder before agent appearance and finally interaction structure are introduced.

Encoder capacity is released progressively: at stage s only the predictor plus the deepest $K_s = \lfloor \kappa_s L \rfloor$ transformer blocks (of depth L) receive gradients, so that at most $8/48$ blocks (ViT-G) and $6/40$ blocks (ViT-g) ever move. The lower blocks and the momentum target network follow the frozen / momentum protocol of the underlying V-JEPA implementation.

Training Objective

FactorJEPA retains the executed V-JEPA objective and adds three auxiliary terms:

$$\boxed{\mathcal{L} = \mathcal{L}_{\text{JEPA}} + \beta\, \mathcal{L}_{\text{probe}} + \gamma\, \mathcal{L}_{\text{motion}} + \lambda\, \mathcal{L}_{\text{drift}}}$$

where $\mathcal{L}_{\text{JEPA}}$ is the masked future-latent loss above, $\mathcal{L}_{\text{probe}}$ is a lightweight multi-task read-off used for checkpoint selection, $\mathcal{L}_{\text{motion}}$ jointly regresses a 13-dimensional optical-flow descriptor and classifies its motion class, and $\mathcal{L}_{\text{drift}}$ is an ℓ_2 anchor to the initialisation $\theta^{(0)}$ that bounds how far the adapted encoder may travel from the pretrained solution.

Predictor Surgery

FactorJEPA is initialised from a pretrained V-JEPA model; the online encoder, momentum target encoder, target construction and masking policy are all retained. Because only the top-K blocks and the predictor are updated, and because $\mathcal{L}_{\text{drift}}$ penalises displacement from $\theta^{(0)}$, adaptation is deliberately *local*: it changes how the predictor is trained, not what it is made of. We refer to this as *predictor surgery*, and evaluate two ablations of it: FactorJEPA-RAW, which applies the identical schedule and budget to raw clips with no factor views, and a no-D_I variant, which omits the interaction stage. Exact stage boundaries, mixture weights, unfreeze fractions, optimiser settings and measured GPU-hours are reported in Appendix C.

Experiments & Evaluation

We evaluate whether FactorJEPA improves the predictive structure of V-JEPA rather than only its downstream semantics, using the four diagnostics defined below.

Evaluator Independence and Held-out Split

To prevent agreement with the pseudo-label generator from being mistaken for improved world modeling, factor targets (Grounding DINO + SAM 3) are constructed only on the training split, and evaluation uses a disjoint held-out split never seen in training, model selection, or tuning. No headline metric reuses FactorJEPA's pseudo-label generator as its evaluator: Future-frame L1 and Mask-ratio slope use only the frozen V-JEPA target encoder, Motion cosine an independently frozen motion estimator, and RGB Agent F1 a frozen detector that touches neither FactorJEPA supervision nor factor-target preprocessing. Full provenance is in Appendix E.

Protocol and Comparisons

We evaluate V-JEPA 2.1 (Mur-Labadia et al. 2026; Assran et al. 2025) at two scales: ViT-G with approximately 2B parameters and ViT-g with approximately 1B parameters. The comparison spans the frozen backbone, conventional full fine-tuning, LP-FT (Kumar et al. 2022), parameter-efficient LoRA, DoRA, Auto-RGN, FactorJEPA-RAW, and full FactorJEPA. FactorJEPA-RAW applies the factorized predictor to raw clips without factor-target supervision, so the attribution chain generic adaptation (LoRA, DoRA, Auto-RGN) $\rightarrow$ FactorJEPA-RAW $\rightarrow$ FactorJEPA separates gains due to parameter adaptation, predictor structure, and structured supervision. Our experiments therefore test a precise claim: whether explicit layout, agent, and interaction structure improves future prediction beyond the adaptation capacity of a monolithic JEPA predictor under a matched protocol.

All methods use the same clips, masking policy, evaluation split, and metric implementations. We report per-encoder test results with 95% BCa bootstrap confidence intervals. In the exported evaluation figures, *surgery* denotes FactorJEPA and *surgery-raw* denotes FactorJEPA-RAW.

Four Diagnostics of Predictive Structure

We report four predictive-structure diagnostics (formal definitions in Appendix E): *Future-frame L1* (↓), the normalized L1 distance between predicted and target future embeddings; *Causal future-block L1* (↓), which forecasts the frozen target encoder's future-half latents from the past half alone, a strictly causal test that uses no factor annotations; *Mask-ratio slope* (↓), the growth in prediction error as visual evidence is progressively masked; and *Motion cosine* (↑), intra- minus inter-class cosine similarity across motion classes.

Decisive Gains over the Frozen World Model

Figure 6 reports FactorJEPA's advantage over the frozen backbone in confidence-interval widths (values beyond $1\times$ indicate separation): it clears the threshold on all four headline diagnostics at both scales, and stays decisive from 2B to 1B. At 1B, FactorJEPA cuts future-frame L1 by 10.6%, causal future-block L1 by 8.9%, and mask-ratio slope by 27.4% against the frozen backbone, and lifts motion-cosine separation to $15.3\times$ its frozen value, each gain exceeding $70\times$ the paired 95% CI. Against the strongest *competing* adaptation (Figure 7) it further separates from *every* fine-tuning rival on Future-frame L1 and Causal future-block L1 (both scales), Mask-ratio slope, and free-running exposure-bias gap (1B), for 24 of 24 paired wins over the six baselines, and it beats the strongest full-scale rival, LoRA, on all four headline diagnostics (+2.8%, +1.6%, +25.2%, and +17.8%).

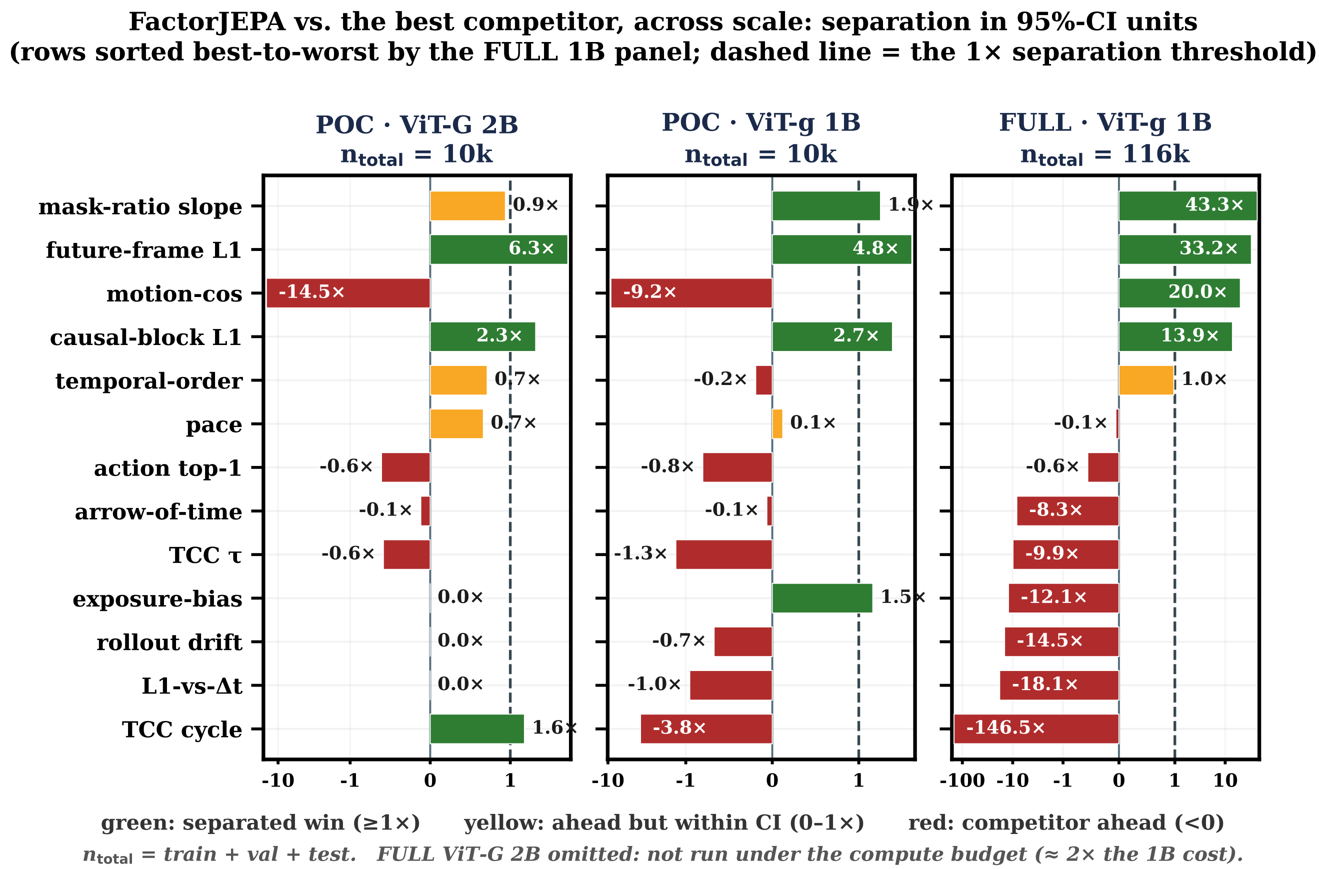

Figure 6: ***FactorJEPA vs. the best competitor, across scale.*** Per-metric advantage in 95%-CI units at POC 2B, POC 1B, and full-scale 1B (green past the dashed 1× line = a separated win). Predictive diagnostics separate and *strengthen* at full scale, where the motion-cosine trade-off also reverses to a win. Full-scale 2B was not run (compute budget).

The gain is structured, not universal: across the fifteen-diagnostic scorecard (supplementary material) FactorJEPA leads the predictive and masking diagnostics while frozen and conventionally tuned models keep motion-linearity and some frame-timing (Misra, Zitnick, and Hebert 2016; Wang, Jiao, and Liu 2020) and temporal-coherence (Wei et al. 2018; Dwibedi et al. 2019) measures. The lone trade-off is Motion cosine, where FactorJEPA improves over frozen but trails the best full-fine-tuning arm: factorization favours accurate, causally structured, occlusion-robust prediction over maximally linear motion readout. Per-metric magnitudes are in Appendix D.

Cross-Scale Replication

The complete ranking of adaptation methods is stable across the 2B and 1B backbones. The per-diagnostic Spearman correlations ρ_k are strongest on the four headline metrics ($\rho = 0.895$ to 0.978; Figure 8), twelve of fifteen diagnostics retain consistent rankings, and only Temporal-order, Teacher-free gap, and Rollout drift fail to transfer (full comparison in the supplementary material). This establishes the 1B model as a faithful lower-cost proxy for screening adaptation choices before full-scale 2B training, and shows the predictive gains and the motion trade-off are properties of the method ordering rather than artifacts of one backbone.

Ablations and Limitations

We isolate each ingredient with matched controls on the same held-out split (paired BCa bootstrap; *separates* means the oriented 95% interval excludes zero); full per-metric detail is in Appendix D. ***The gain is predictor surgery, not the factor targets:*** FactorJEPA-RAW (identical schedule and unfreeze budget, no factor supervision) is not beaten by full FactorJEPA on future-frame at either scale, so the closing of the frozen-to-adapted gap comes from the surgical adaptation, not the factor-view supervision layered on top. ***Updating the encoder, not only the head, drives the wins:*** head-only adaptation loses on future-frame, causal future-block, and mask-ratio (each separated) but wins back motion-cosine, rollout, and exposure-bias. ***The interaction stage adds no measurable value:*** removing D_I (no-D_I) leaves every predictive diagnostic statistically unchanged. ***Weight merging hurts:*** a WiSE-FT (Wortsman et al. 2022) interpolation (mixing fractions 0.3, 0.5, 0.7) loses to the un-merged model on future-frame and motion-cosine at both scales.

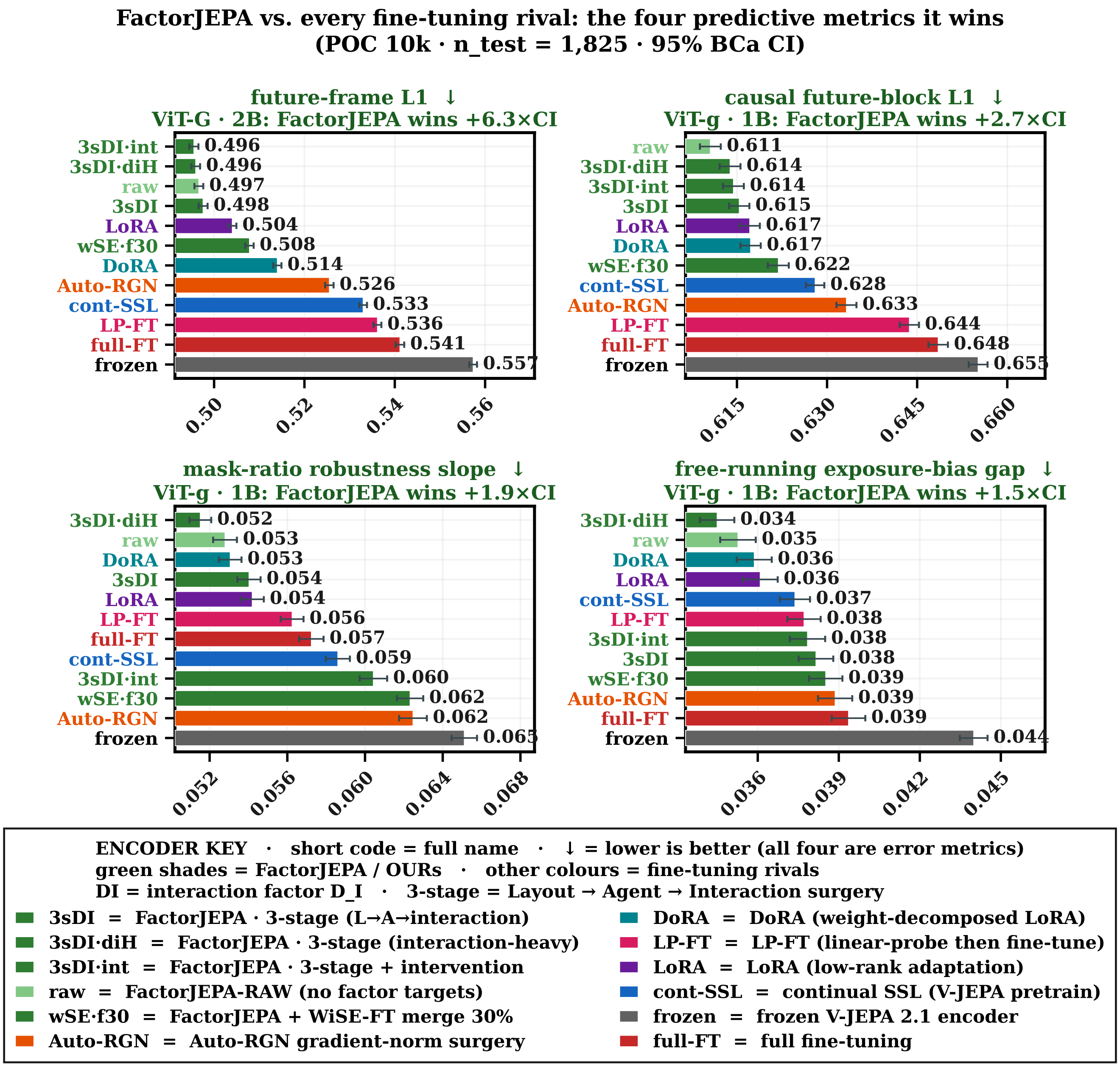

Figure 7: ***FactorJEPA outperforms every fine-tuning rival on the four predictive metrics it wins outright*** (Future-frame L1 at 2B; Causal future-block L1, Mask-ratio slope, exposure-bias gap at 1B). Bars show held-out score ± 95% BCa CI; every rival is named in the key. Full 15-diagnostic scorecard in the supplementary material.

Limitations. Three controls bound our claims. A compute-matched control (plain pretraining for the same number of steps as the surgery) is declared but not run, so we cannot fully separate the surgical schedule from additional optimisation; head-only pretraining and the continual-learning baselines (CaSSLe (Fini et al. 2022), EWC (Kirkpatrick et al. 2017)) were trained but not evaluated on the suite; and the method is a precondition on the base checkpoint, as on V-JEPA 2.0 the same recipe does not improve over the frozen encoder, so the gains are specific to V-JEPA 2.1. At the 10k screening scale factorization also concedes maximally linear motion readout to full fine-tuning, a trade-off that reverses only at full (86,831-video) scale.

Conclusion

We studied world modeling in populous, crowded, and chaotic Global South cities, where heterogeneity, occlusion, and mixed-traffic negotiation challenge conventional JEPA predictors. We introduced DENSEWORLD: 276 hours of drive-through, walk-through, and aerial video collected

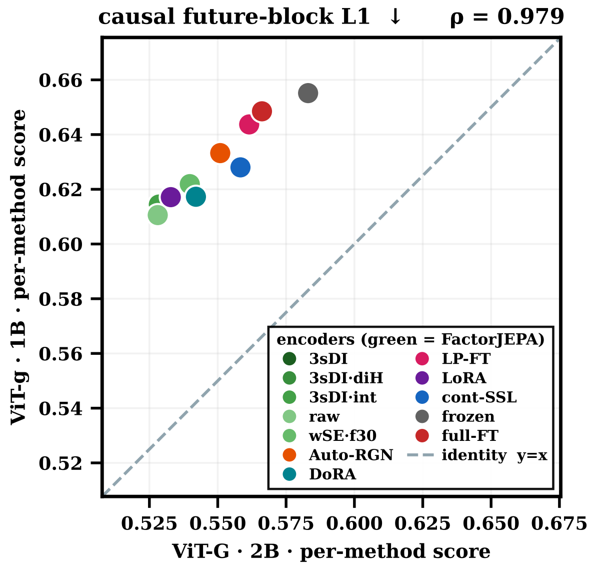

Figure 8: ***Method rankings survive the 2B→1B backbone change.*** Per-method *Causal future-block L1* at both scales (Spearman $\rho = 0.978$), making the 1B backbone a low-cost proxy for 2B. Full 15-panel grid in the supplementary material.

across 22 cities. Building on this benchmark, we proposed FactorJEPA, which makes world structure a first-class predictive primitive through *predictor surgery*: a three-stage curriculum that adapts the predictor over automatically separated *layout*, *agent*, and *interaction* views. Across 2B and 1B backbones, FactorJEPA delivers (i) more accurate future-latent prediction, (ii) stronger causal future-block structure, and (iii) greater robustness under reduced visual evidence, with method rankings reproducing across scale at $\rho = 0.895$ to 0.978. The results also expose a trade-off between structured prediction and maximally linear motion readout at the 10k screening scale, which reverses at full scale (86,831 factor-annotated videos). Together, these contributions move JEPA world models from *monolithic predictors* toward *structured, factor-aware models* of complex real-world dynamics.

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